

# Scale Fixes Local Errors First: Differential Scaling of Chess Transformer Failures

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## Declarations

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**Data availability statement.** The analysis code, every result file cited by the manuscript, and the macro provenance record are available at <https://github.com/MohitPratapSinghRathore/derived-total>. Trained checkpoints are available from the corresponding author on reasonable request, being too large to distribute with the repository. External checkpoints are public.

**AI usage declaration.** During the preparation of this manuscript the authors used a generative AI tool to assist with language editing and refinement of the text, and with writing analysis code. The authors reviewed and edited all output and take full responsibility for the content of the paper. Generative AI was not used to generate or analyse the study’s data, which were produced entirely by the measurement pipeline described in the paper.

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## Contributor biographies

**Mohit Pratap Singh Rathore** is a researcher at Ovigo Private Limited, where his work concerns the measurement of internal state in sequence models and the evaluation of long-horizon failure.

**Gunveer Sinh Kalsi** is a researcher at Ovigo Private Limited, working on model evaluation infrastructure and the reproducibility of measured claims.

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